

Clinical Variant Interpretation Report

Patient/Sample ID: Demo_Patient_001

Report Date: 2026-09-06

Disclaimer: Generated for a portfolio/learning project using public reference cell-line data. Not a clinical diagnostic report.

Pathogenic / Likely Pathogenic variants found: 4

Gene Symbol	Protein Change	Variant Classification	Inferred Activity	Frequency
TP53	p.R175H	Pathogenic	Loss-of-function	0.2767
BRAF	p.V600E	Pathogenic (Tier I)	Activating (Gain-of-function)	0.2958
KRAS	p.G12C	Pathogenic (Tier I)	Activating (Gain-of-function)	0.2708
BRCA1	p.S563fs	Likely Pathogenic	Loss-of-function	0.32

Gene Info & Drug-Response Notes

BRAF: V600E mutation predicts response to BRAF inhibitors (vemurafenib, dabrafenib) + MEK inhibitor.

BRCA1: Loss-of-function predicts sensitivity to PARP inhibitors (olaparib, talazoparib).

KRAS: KRAS mutation predicts resistance to anti-EGFR therapy (cetuximab, panitumumab).

TP53: No approved targeted therapy directed at TP53 itself; consider trial eligibility.

Full variant table (4 total variants) is in the accompanying Excel file.